BRAINSTORMING AND IDEATION

1. PROJECT TITLE

Auto Ticket Classification using Flow Designer

2. PROJECT OVERVIEW

The proposed project focuses on automating the classification of IT support tickets using Flow Designer in the ServiceNow platform.

The school IT helpdesk receives multiple incident requests daily from students and teachers, such as Wi-Fi issues, projector failures, password problems, and slow computers. Currently, IT staff manually review each request and assign an appropriate category and subcategory.

This manual process is time-consuming and may result in inconsistent ticket classification. The proposed system automates the ticket classification process by analyzing keywords in the Short Description and Description fields.

Based on the identified issue, the system automatically assigns the appropriate Category and Subcategory. The solution is implemented using Flow Designer, providing a no-code and easily maintainable approach for automating ticket classification.

3. PROBLEM IDENTIFICATION

The school IT helpdesk receives multiple support requests every day from students and teachers related to network connectivity, hardware failures, account access problems, and system performance issues.

Currently, IT staff manually review each ticket to:

Identify the issue type.
Select the correct category and subcategory.
Process the ticket based on the identified issue.
Inform the caller about ticket creation.

This manual process is time-consuming, error-prone, and not scalable as the number of support requests increases.

Without an automated classification system, IT staff need to spend additional time reviewing and categorizing each ticket. Therefore, an automated solution is required to classify IT tickets efficiently and consistently.

4. PROPOSED SOLUTION

The proposed solution uses ServiceNow Flow Designer to automatically classify IT support tickets based on keywords found in the ticket description.

When a student or teacher creates a new IT support ticket, the system analyzes the Short Description and identifies predefined keywords related to the reported issue.

Based on the identified keyword, the system automatically assigns the appropriate Category and Subcategory.

The classification logic includes:

Issue	Category	Subcategory
Wi-Fi / Network	Network	Wi-Fi
Projector	Hardware	Projector
Password / Login	Access	Forgot Password
Slow / Hanging	Performance	Slow Computer

For example, if the Short Description contains the keyword Wi-Fi, the system automatically assigns Category as Network and Subcategory as Wi-Fi.

If the ticket contains a projector-related issue, the system assigns Category as Hardware and Subcategory as Projector.

Similarly, password or login-related issues are classified under Access, while slow or hanging computer issues are classified under Performance.

An automated email notification is also sent to the caller after the ticket is created.

5. PROJECT OBJECTIVES

The primary objectives of the project are:

1. To automatically classify IT tickets at the time of creation.
2. To reduce manual effort for IT staff.
3. To improve ticket routing efficiency.
4. To automatically assign Category and Subcategory.
5. To implement dependent choice logic between Category and Subcategory.
6. To send an automated email notification to the caller upon ticket creation.
7. To store ticket information in a structured and standardized format.
8. To implement a no-code and easily maintainable solution.
9. To provide a scalable solution for handling increasing numbers of IT support requests.

6. PROPOSED SYSTEM

The proposed system consists of the following major components:

6.1 Ticket Creation

A student or teacher creates a new IT support ticket by providing the required information, including the Caller and Short Description of the issue.

6.2 Ticket Information

The ticket stores important information required for managing IT support requests.

The major fields include:

Number
Caller
Category
Subcategory
Short Description
Description
State
Assigned Group
Assigned To

6.3 Automatic Ticket Classification

After a ticket is created, Flow Designer evaluates the Short Description and identifies predefined keywords.

Based on the identified keyword, the system automatically determines and assigns the appropriate Category and Subcategory.

6.4 Category and Subcategory Dependency

The system implements a dependency relationship between Category and Subcategory.

The dependency is configured as follows:

Wi-Fi Network
Projector Hardware
Forgot Password Access
Slow Computer Performance

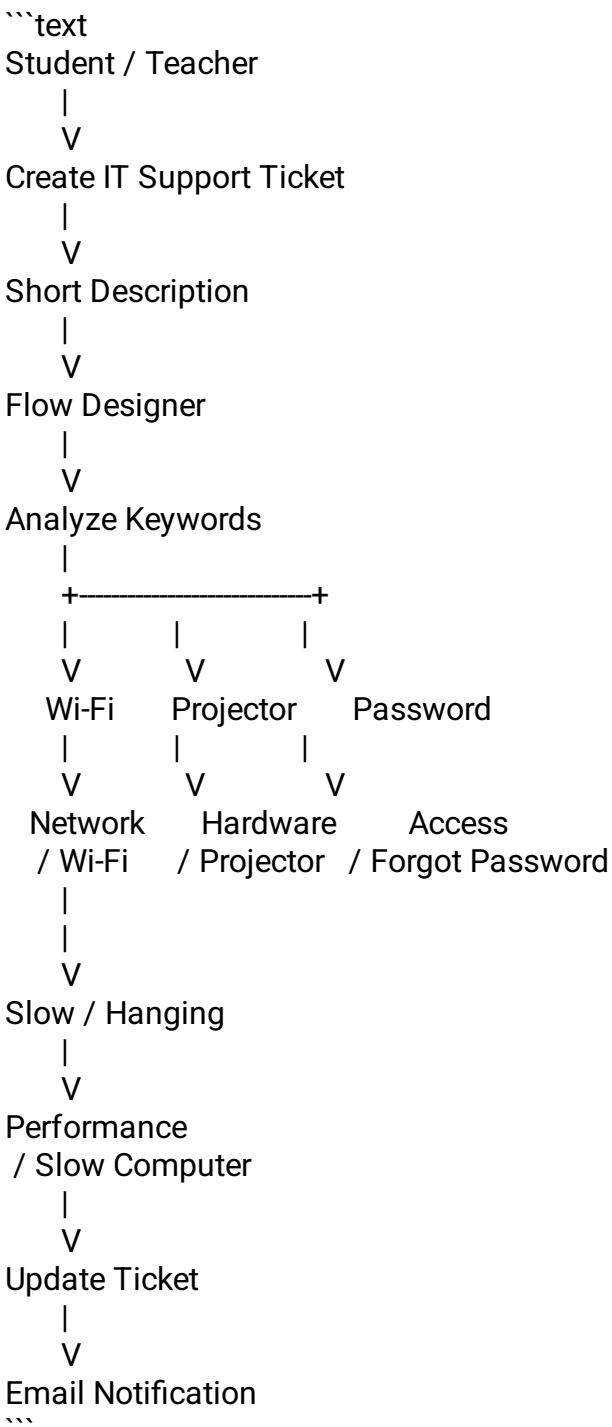
This ensures that the Subcategory values are associated with the appropriate Category.

6.5 Email Notification

After the ticket is classified, an automated email notification is sent to the caller to confirm that the support request has been submitted.

7. AUTOMATED CLASSIFICATION MODEL

The overall ticket classification process is represented below:



The Flow Designer evaluates the ticket description and automatically updates the corresponding Category and Subcategory.

8. AUTOMATION WORKFLOW

The proposed automation follows a systematic workflow:

1. A student or teacher creates a new IT support ticket.
2. The caller and Short Description are entered.
3. The ticket is submitted.
4. Flow Designer is triggered when the record is created.
5. The system checks the Short Description for predefined keywords.
6. The appropriate Category is identified.
7. The corresponding Subcategory is assigned.
8. The ticket record is automatically updated.
9. An email notification is sent to the caller.

This automation eliminates the need for IT staff to manually classify every incoming ticket.

9. BUSINESS USE CASE

The main business use case of the project is to automatically classify IT support tickets received by the school IT helpdesk.

The system must:

1. Automatically classify IT tickets based on the issue description.
2. Assign both Category and Subcategory without manual intervention.
3. Support dependent choice logic between Category and Subcategory.
4. Send an automated email notification to the caller upon ticket creation.
5. Store ticket information in a structured and standardized format.
6. Ensure easy maintenance and future scalability.

The proposed system helps the IT helpdesk manage support requests in a consistent and automated manner.

10. PROJECT SCOPE

The scope of the project includes:

Creating the required table to store ticket records.
Creating the required fields.
Configuring Category and Subcategory choices.
Implementing dependency between Category and Subcategory.
Creating the Flow Designer automation.
Configuring keyword-based ticket classification.
Automatically updating Category and Subcategory.
Configuring automated email notification.
Testing different types of IT support tickets.
Validating the classification results.
Documenting the implementation and results.

The project is focused on demonstrating automated IT ticket classification using ServiceNow Flow Designer.

11. EXPECTED OUTCOME

The expected outcome is a functional ServiceNow automation system in which IT support tickets are automatically classified using Flow Designer.

The completed implementation should demonstrate that:

IT tickets can be automatically classified.
Category and Subcategory can be assigned based on issue keywords.
Dependent choice fields can maintain consistent ticket information.
Flow Designer can automate the classification process.
Email notifications can be generated automatically.
Manual effort required for ticket classification can be reduced.
The solution can be maintained and extended in the future.

12. BENEFITS

The proposed solution provides the following benefits:

Reduced Manual Effort: IT staff do not need to manually classify every incoming ticket.
Improved Classification: Tickets are classified according to predefined conditions.
Faster Processing: Automatic classification reduces the time required to process support requests.
Consistent Data: Category and Subcategory values are assigned in a standardized manner.
Automated Communication: Email notifications are automatically sent to callers.
Easy Maintenance: Flow Designer provides a simple and maintainable automation approach.
Scalability: The system can be extended to support additional ticket types and conditions.
Improved Efficiency: The overall ticket handling process becomes more systematic and efficient.

13. FUTURE ENHANCEMENTS

The project can be extended with additional features, including:

Adding more ticket categories and subcategories.
Supporting additional keywords and issue types.
Automatic assignment of tickets to support groups.
Automatic assignment of tickets to individual agents.
SLA-based ticket management.

Automatic priority determination.
Advanced reporting and dashboard capabilities.
Integration with additional ServiceNow applications.
Predictive intelligence for advanced ticket classification.
Automated escalation for unresolved tickets.

These enhancements can extend the basic classification system into a more comprehensive IT service management solution.

14. CONCLUSION

The Auto Ticket Classification using Flow Designer project provides an automated solution for classifying IT support tickets in a school environment.

The project addresses the problem of manually reviewing and categorizing support requests by using ServiceNow Flow Designer to analyze ticket descriptions and automatically assign the appropriate Category and Subcategory.

The implementation includes dependent choice logic, automated ticket updates, and email notification to the caller. The proposed solution reduces manual effort, improves classification efficiency, and provides a structured approach for handling IT support requests.

Overall, the project demonstrates how ServiceNow Flow Designer can be effectively used to automate a practical IT helpdesk process using a no-code approach. The solution can be further enhanced in the future by adding assignment automation, SLA tracking, advanced reporting, and predictive intelligence.